

Exercise 2

EXERCISE 2 · SQL AGGREGATES & OPERATORS

The students Table

SQL Editor

```
21 -- Display Informations from the table
22 SELECT *
```

Table

id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

15 rows 13.314s runtime See performance (1) Checked for improvements Refreshed 1m ago

Q1. List all distinct departments in the students table.
EXPECTED COLUMNS: department.

SQL Editor

```
18 --Q1 List all distinct departments in the students table.
19 SELECT DISTINCT department
20 FROM brightlearn_exc2.courses;
```

Table

department
IT
HR
Finance

Q2 Get the average age of students per department.

EXPECTED COLUMNS: department avg_age

Workspace

SQL Editor

```
22 --Q2 Get the average age of students per department.
23 SELECT department, AVG(age) as avg_age
24 FROM brightlearn_exc2.courses
25 GROUP BY department;
```

Table

department	avg_age
IT	20.5
HR	22
Finance	23

Q3. Show departments with more than 1 student.
EXPECTED COLUMNS: department student_count

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Home, Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL and SQL Editor. The main editor area displays a SQL query:

```
--03 Show departments with more than 1 student
SELECT department, COUNT(*) as num_students
FROM brightlearn_exc2.courses
GROUP BY department
HAVING num_students > 1;
```

Below the query editor is a table viewer showing the results of the query:

	department	num_students
1	IT	2
2	HR	2

Q4. Get all students whose age is between 21 and 23.
EXPECTED COLUMNS: student_id name age department

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL, SQL Editor, and Queries. The main editor area displays a SQL query:

```
--04 Get all students whose age is between 21 and 23.
SELECT *
FROM brightlearn_exc2.courses
WHERE age BETWEEN 21 AND 23;
```

Below the query editor is a table viewer showing the results of the query:

	id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Q5. List all students in the IT or HR department who are older than 21.
EXPECTED COLUMNS: student_id name age department

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL and SQL Editor. The main editor area displays a SQL query:

```
-- 05 List all students in the IT or HR department who are older than 21.
SELECT *
FROM brightlearn_exc2.courses
WHERE (department = 'IT' OR department = 'HR') AND age > 21;
```

Below the query editor is a table viewer showing the results of the query:

	id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

The courses Table

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

```
22 -- Display Informationa from the table
23 SELECT *
24 FROM brightlearn_exc2.academy.courses;
25
```

Add parameter

	id	name	age	department
1	1	Alice	20	IT
2	2	Bob	22	HR
3	3	Charlie	21	IT
4	4	Diana	23	Finance
5	5	Eve	22	HR

06 Show total credits per department, only for departments with more than 5 total credits.
EXPECTED COLUMNS: department total_credits

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

```
66
67 --06 Show total credits per department, only for departments with more than 5 total credits.
68 SELECT department AS department, SUM(credits) AS total_credits
69 FROM courses
70 GROUP BY department
71 HAVING total_credits > 5;
72
```

Add parameter

	department	total_credits
1	IT	11

Q7. List all courses that do not have 4 credits.
EXPECTED COLUMNS: course_id course_name department credits

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

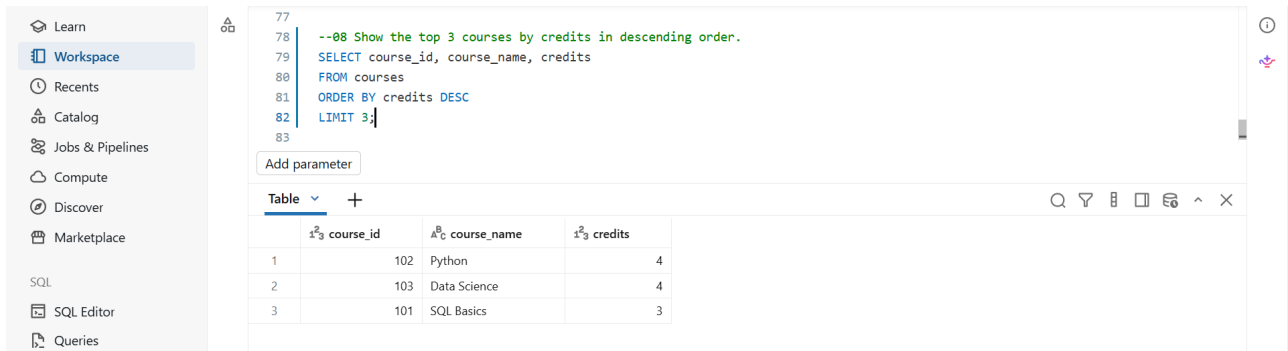
Queries

```
71 HAVING total_credits > 5;
72
73 --07 List all courses that do not have 4 credits.
74 SELECT *
75 FROM courses
76 WHERE credits != 4;
77
```

Add parameter

	course_id	course_name	department	credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

Q8. Show the top 3 courses by credits in descending order.
EXPECTED COLUMNS: course_id course_name credits



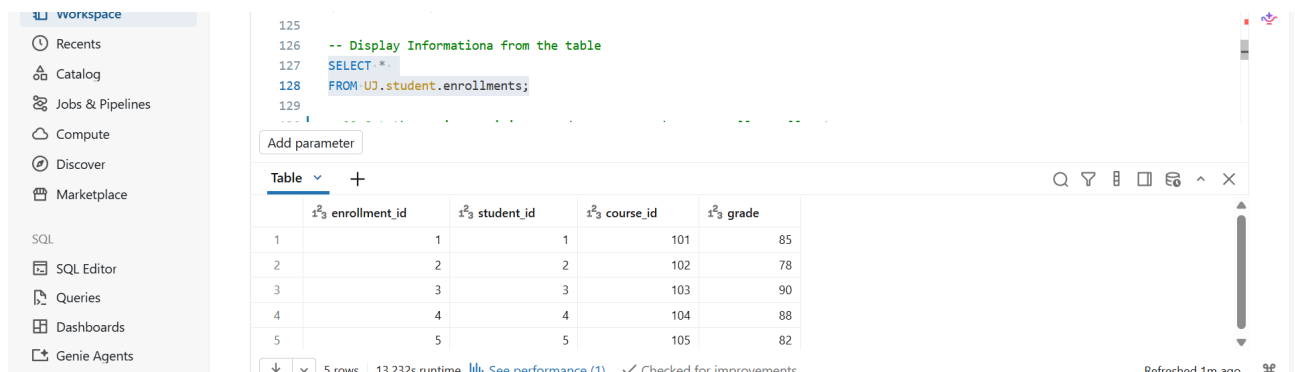
The screenshot shows a SQL workspace interface. On the left is a sidebar with navigation options: Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor and Queries. The main area displays a SQL query editor with the following code:

```
77
78 --Q8 Show the top 3 courses by credits in descending order.
79 SELECT course_id, course_name, credits
80 FROM courses
81 ORDER BY credits DESC
82 LIMIT 3;
83
```

Below the editor is a table viewer showing the results of the query:

	course_id	course_name	credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

The enrollments Table



The screenshot shows a SQL workspace interface. On the left is a sidebar with navigation options: workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor, Queries, Dashboards, and Genie Agents. The main area displays a SQL query editor with the following code:

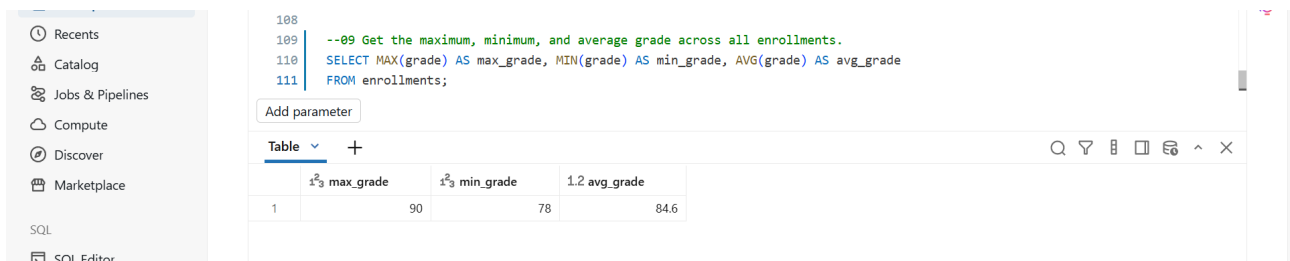
```
125
126 -- Display Informationa from the table
127 SELECT *
128 FROM UJ.student.enrollments;
129
```

Below the editor is a table viewer showing the results of the query:

	enrollment_id	student_id	course_id	grade
1	1	1	101	85
2	2	2	102	78
3	3	3	103	90
4	4	4	104	88
5	5	5	105	82

At the bottom of the table viewer, it indicates "5 rows" and "13.232e runtime".

Q9. Get the maximum, minimum, and average grade across all enrollments.
EXPECTED COLUMNS: max_grade min_grade avg_grade



The screenshot shows a SQL workspace interface. On the left is a sidebar with navigation options: Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor. The main area displays a SQL query editor with the following code:

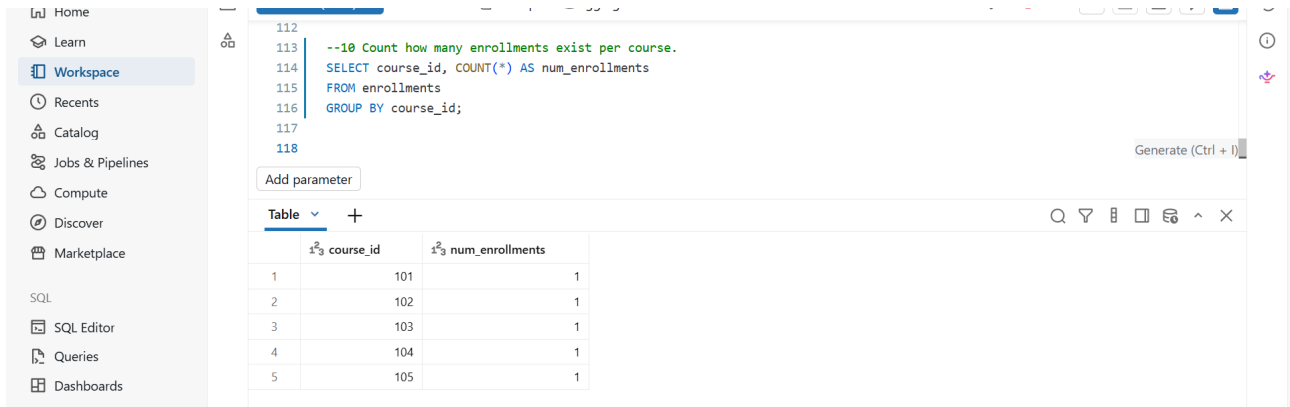
```
108
109 --Q9 Get the maximum, minimum, and average grade across all enrollments.
110 SELECT MAX(grade) AS max_grade, MIN(grade) AS min_grade, AVG(grade) AS avg_grade
111 FROM enrollments;
```

Below the editor is a table viewer showing the results of the query:

	max_grade	min_grade	avg_grade
1	90	78	84.6

Q10. Count how many enrollments exist per course.

EXPECTED COLUMNS: course_id enrollment_count



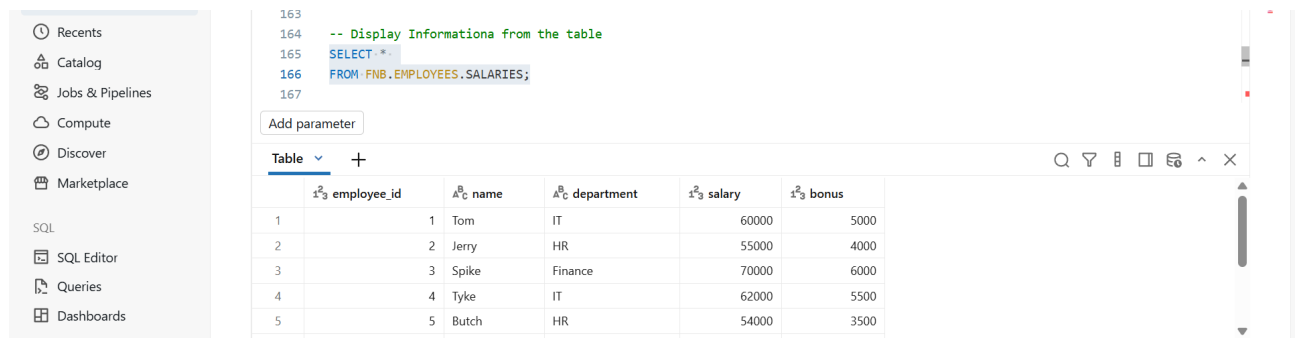
The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Home, Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor, Queries, and Dashboards. The main editor area contains a SQL query:

```
112
113 --Q10 Count how many enrollments exist per course.
114 SELECT course_id, COUNT(*) AS num_enrollments
115 FROM enrollments
116 GROUP BY course_id;
117
118
```

Below the query editor is a "Table" view showing the results of the query:

	course_id	num_enrollments
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

The salaries Table



The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor, Queries, and Dashboards. The main editor area contains a SQL query:

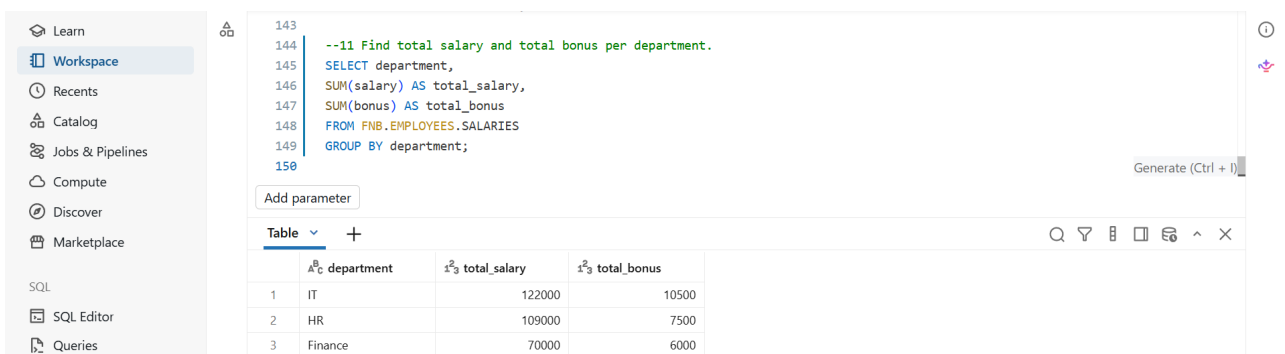
```
163
164 -- Display Informations from the table
165 SELECT *
166 FROM FNB.EMPLOYEES.SALARIES;
167
```

Below the query editor is a "Table" view showing the results of the query:

	employee_id	name	department	salary	bonus
1	1	Tom	IT	60000	5000
2	2	Jerry	HR	55000	4000
3	3	Spike	Finance	70000	6000
4	4	Tyke	IT	62000	5500
5	5	Butch	HR	54000	3500

Q11. Find total salary and total bonus per department.

EXPECTED COLUMNS: department total_salary total_bonus



The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related options: SQL Editor, Queries, and Dashboards. The main editor area contains a SQL query:

```
143
144 --Q11 Find total salary and total bonus per department.
145 SELECT department,
146 SUM(salary) AS total_salary,
147 SUM(bonus) AS total_bonus
148 FROM FNB.EMPLOYEES.SALARIES
149 GROUP BY department;
150
```

Below the query editor is a "Table" view showing the results of the query:

	department	total_salary	total_bonus
1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

Q12. Show departments where average salary is above 55,000.

EXPECTED COLUMNS: department avg_salary

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related tools: SQL Editor, Queries, Dashboards, and Genie Agents. The main editor displays a SQL query for Q12: `-- 12 Show departments where average salary is above 55,000. SELECT department, AVG(salary) AS avg_salary FROM FNB.EMPLOYEES.SALARIES GROUP BY department HAVING AVG(salary) > 55000;`. Below the query editor is a table viewer showing the results of the query. The table has columns: employee_id, name, department, salary, and bonus. The results are as follows:

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

Q13. List employees whose salary plus bonus is greater than 60,000.

EXPECTED COLUMNS: employee_id name salary bonus total_compensation

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related tools: SQL Editor, Queries, and Dashboards. The main editor displays a SQL query for Q13: `--Q13 List employees whose salary plus bonus is greater than 60,000. Select employee_id, name, salary, bonus, salary + bonus AS total_compensation FROM FNB.EMPLOYEES.SALARIES WHERE salary + bonus > 60000;`. Below the query editor is a table viewer showing the results of the query. The table has columns: employee_id, name, salary, bonus, and total_compensation. The results are as follows:

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
2	Spike	70000	6000	76000
3	Tyke	62000	5500	67500

The projects Table

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are SQL-related tools: SQL Editor, Queries, and Dashboards. The main editor displays a SQL query: `-- Create schema Create schema if not exists african.bank;`. Below the query editor is a table viewer showing the results of the query. The table has columns: project_id, project_name, department, and budget. The results are as follows:

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

Q14. Show total and average budget per department. Only include departments with average budget above 70,000.
EXPECTED COLUMNS: department total_budget avg_budget.

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below this is a section for SQL, containing SQL Editor, Queries, and Dashboards. The main editor area displays a SQL query for Q14:

```

191
192 Q14 Show total and average budget per department. Only include departments with average budget
193 above 70,000.
194 SELECT department,
195 SUM(budget) AS total_budget,
196 AVG(budget) AS avg_budget
197 FROM african.bank.projects
198 GROUP BY department
199 HAVING avg_budget > 70000;
  
```

Below the query editor is a table view showing the results:

	department	total_budget	avg_budget
1	IT	270000	135000
2	Finance	80000	80000

Q15. List all projects with budgets between 50,000 and 120,000, excluding the Marketing department.
EXPECTED COLUMNS: project_id project_name department budget

The screenshot shows a SQL IDE interface. On the left is a sidebar with navigation options: Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below this is a section for SQL, containing SQL Editor, Queries, and Dashboards. The main editor area displays a SQL query for Q15:

```

200
201 --Q15 List all projects with budgets between 50,000 and 120,000, excluding the Marketing department.
202 SELECT *
203 FROM african.bank.projects
204 WHERE budget BETWEEN 50000 AND 120000 AND TRIM(department) != 'Marketing';
  
```

Below the query editor is a table view showing the results:

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000

-----EXERCISE 2 · SQL AGGREGATES & OPERATOR-----

